

Multi-Omics Predictive Modelling for Breast Cancer Relapse

Benchmarking against clinical staging, with external validation · 3-year relapse endpoint
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Abstract

Headline result. On identical outer folds, the clinical baseline performed best. Model A (clinical: age, stage, PAM50) reached AUC **0.779** [0.762-0.796]; Model B (XGBoost on clinical + expression + methylation) reached **0.669** [0.648-0.690]; Model C (pathway-informed neural network) reached **0.613** [0.593-0.633]. Adding ~19,000 omics features did not improve on a two-variable clinical model at this sample size.

Cohort. n = 367 patients with matched expression, methylation and a usable label; 68 relapse-positive (18.5%). 586 patients were excluded for censoring before the 3-year horizon.

External validation. METABRIC n = 1902 (359 positive). Best expression-only AUC **0.6321** (panel 20, logreg_I2). METABRIC has no matched 450k methylation, so the methylation half of the signature could not be tested externally.

What this report does and does not claim

This is a negative-leaning result reported as found. The pipeline was pre-specified: feature selection is nested inside every cross-validation split, models are compared on identical fold indices, and no model was retuned after seeing its comparison. Where the data could not support the planned analysis, the deviation is stated rather than worked around.

1. Data and cohort construction

1.1 Sources (Gate 0)

Dataset	Rows	Cols	Size
tcga_expression	60,660	1,227	138.5 MB
tcga_methylation	486,427	894	2.99 GB
tcga_clinical	1,255	85	0.3 MB
tcga_survival	1,232	4	0.0 MB
tcga_cdr	12,591	34	2.4 MB
tcga_subtype	7,734	9	0.0 MB
methylation_manifest	486,428	33	64.0 MB
mcp_genes	111	4	0.0 MB
encode_gtf	3,014,859	1	44.5 MB
metabric_expr	20,603	1,982	689.3 MB
metabric_patient	2,513	24	0.4 MB
metabric_sample	2,513	13	0.3 MB

Full URLs and md5 checksums are in [results/tables/gate0_data_manifest.csv](#).

1.2 Label definition

Relapse is defined from the Progression-Free Interval (PFI) in the TCGA-CDR curated endpoints (Liu et al., Cell 2018), at a 1095-day (3-year) horizon:

PFI event	PFI time	Label
1	$\leq 1095d$	positive (relapsed)
0	$\geq 1095d$	negative (relapse-free)
0	$< 1095d$	EXCLUDED - censored early, status unknown
1	$> 1095d$	negative

The third row is the one that is easy to get wrong. A patient censored at two years is not a negative; their five-year status is unknown. Excluding them is correct and expensive: it is the single largest filter in this pipeline.

1.3 Horizon choice (deviation from spec)

The build spec targeted a 5-year horizon and expected $n = 550-750$ with 15-25% positives. Measured over the patient universe with matched expression AND methylation (782 patients), no horizon satisfies both the $n \geq 400$ and positives ≥ 60 floors simultaneously. The binding constraint is the multi-omics intersection, not the horizon.

Horizon	n	Positives	Prevalence	Excluded (early censor)	Meets $n \geq 400$	Meets $pos \geq 60$
2 y	460	49	10.7%	317 (41%)	yes	no
3 y	367	68	18.5%	410 (53%)	no	yes
4 y	297	80	26.9%	480 (62%)	no	yes
5 y	254	94	37.0%	523 (67%)	no	yes

Decision: the 3-year horizon is used as primary. It clears the positives floor and is the only horizon whose prevalence lands inside the expected 15-25% band; it misses the $n \geq 400$ floor. The 5-year analysis is retained in full as a sensitivity run ([results/tables_5y/](#)).

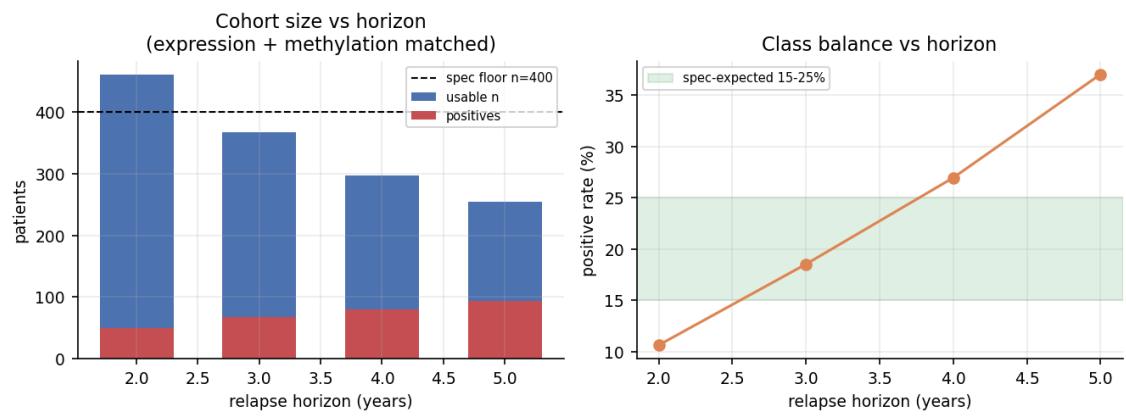


Figure 1. Cohort size and class balance against relapse horizon.

1.4 Filtering accounting (Gate 1)

Step	Unit	In	Out	Dropped
TCGA-CDR rows (all cancer types)	samples	12,591	12,591	
restrict to BRCA	samples	12,591	1,236	11,355
drop TCGA-redacted patients	samples	1,236	1,230	6
de-duplicate to one row per patient	samples	1,230	1,092	138
drop patients with missing PFI or PFI.time	samples	1,092	1,091	1
EXCLUDE censored before 1095d (status unknown)	samples	1,091	505	586
positives (relapse <= 5y)	patients	89	89	
negatives (relapse-free through 5y)	patients	416	416	
expression: raw genes	genes	60,660	60,660	
expression: raw samples (aliquots)	aliquots	1,226	1,226	
expression: keep primary tumour (sample type 01)	aliquots	1,226	1,106	120
expression: collapse aliquots to one per patient	aliquots	1,106	1,095	11
expression: drop chrY PAR_Y duplicate accessions	genes	60,660	60,616	44
expression: genes mappable to GENCODE v36	genes	60,616	60,616	
expression: keep protein_coding	genes	60,616	19,944	40,672
expression: collapse duplicate gene symbols (mean)	genes	19,944	19,938	6
expression: drop zero-variance genes	genes	19,938	19,516	422
expression: drop bottom 50% by variance	genes	19,516	9,758	9,758
450k manifest probes	probes	486,428	486,428	
manifest: drop chrX/chrY probes	probes	486,428	474,780	11,648
manifest: probes with a RefGene annotation	probes	356,119	356,119	
manifest: promoter-associated probes (1stExon,5'UTR,TSS1500,TSS200)	probes	356,119	192,998	163,121
methylation: probes in file	probes	486,427	192,998	293,429
methylation: keep primary tumour (sample type 01)	aliquots	893	791	102
methylation: collapse aliquots to one per patient	aliquots	791	784	7
methylation: drop probes with >10% NA	probes	192,998	166,054	26,944
methylation: drop samples with >10% NA	patients	784	784	
methylation: aggregate promoter probes to gene mean beta	genes	181,896	19,146	162,750
methylation: drop bottom 50% by variance	genes	19,146	9,573	9,573
patients with a usable label	patients	505	505	
patients with expression	patients	1,095	1,095	
patients with methylation	patients	784	784	
INTERSECT label AND expression AND methylation	patients	505	367	138
genes shared by both modalities	genes	5,669	5,669	

Final cohort: n = 367, 68 positive / 299 negative (18.5% prevalence), 9758 expression features, 9573 methylation features, 5669 genes measured by both modalities.

2. Model comparison

All three models are evaluated on the SAME outer fold indices (5-fold x 10 repeats = 50 folds), loaded from a written fold file rather than regenerated from a seed, so the comparison is genuinely paired.

Model	Description	AUC (95% CI)	AP (95% CI)
A	Logistic regression, clinical only (age, stage, PAM50)	0.779 [0.762, 0.796]	0.546 [0.516, 0.576]
B	XGBoost, clinical + expression + methylation	0.669 [0.648, 0.690]	0.364 [0.338, 0.389]
C	Pathway-informed neural network (P-NET style)	0.613 [0.593, 0.633]	0.308 [0.288, 0.329]

2.1 Paired tests on identical folds

Comparison	Mean AUC difference	Paired t p	Wilcoxon p	Folds where 2nd wins
A vs B	-0.1098	2.99e-11	1.02e-10	8/50
A vs C	-0.1656	9.18e-18	1.24e-14	2/50
B vs C	-0.0558	8.17e-11	2.81e-10	4/50

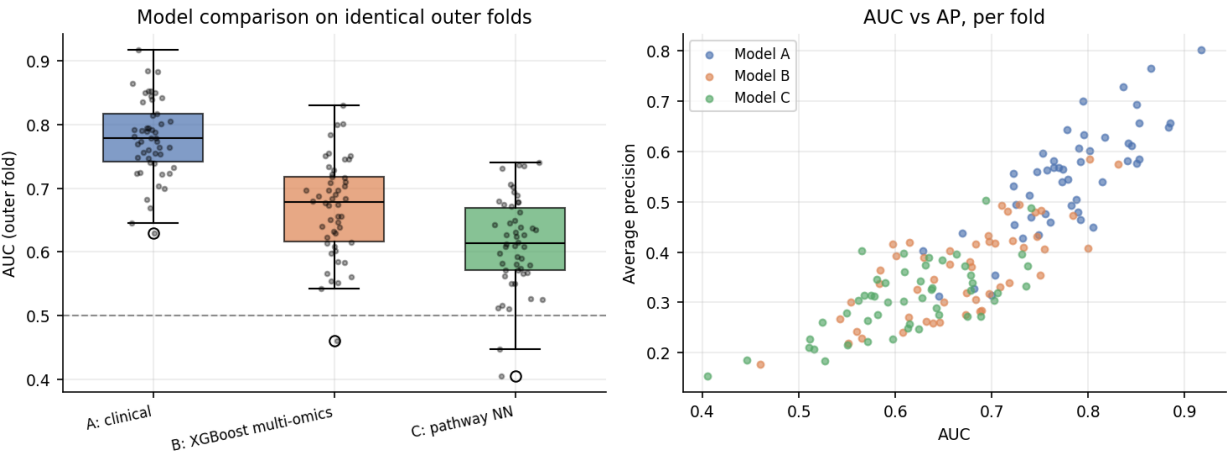


Figure 2. Per-fold AUC by model on identical outer folds.

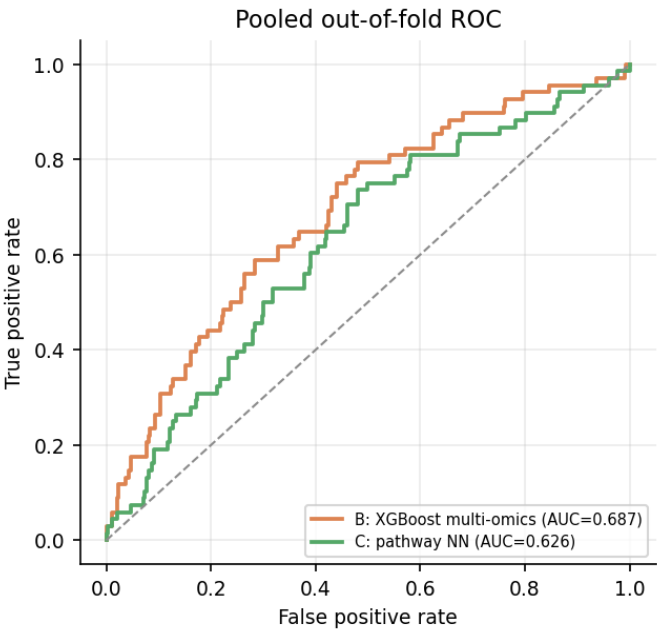


Figure 3. Pooled out-of-fold ROC curves.

2.2 Gate 2 diagnostic: why is the clinical baseline strong?

Model A exceeded the 0.60-0.70 band the spec expects for clinical-only BRCA models, so it was investigated before proceeding. It is not leakage: PAM50 (which is expression-derived, and so arguably should not sit in a 'clinical' baseline at all) contributes almost nothing, and the signal is carried by pathologic stage.

Variant	AUC (95% CI)	AP
A_full (age+stage+PAM50)	0.779 [0.762, 0.796]	0.546
A_no_PAM50 (age+stage only, TRUE clinical)	0.769 [0.752, 0.786]	0.544
A_PAM50_only	0.571 [0.546, 0.596]	0.239
A_age_only	0.637 [0.615, 0.659]	0.365
A_stage_only	0.718 [0.704, 0.732]	0.384
A_full_excl_stageIV	0.744 [0.723, 0.765]	0.454
A_no_PAM50_excl_stageIV	0.733 [0.713, 0.753]	0.447

Stage composition of the cohort:

Stage	n	Relapse rate
missing	10	30.0%
1	62	3.2%
2	197	13.2%
3	88	31.8%
4	10	90.0%

The stage gradient is steep, and the landmark labelling rule sharpens it further: requiring survivors to have full follow-up preferentially retains lower-stage patients as negatives, while early events remain as positives. The strong clinical baseline is a real property of this selected cohort, not an artefact of the code.

3. Feature ranking (Gate 3)

#	Feature	Block	Mean SHAP	Fold selection
1	TRIML2	methylation	0.4623	2%
2	stage	clinical	0.3446	100%
3	BAK1	methylation	0.3298	2%
4	PC	expression	0.2956	2%
5	MIR1284	methylation	0.2723	100%
6	TNFSF11	expression	0.2638	98%
7	FLT1	expression	0.2603	2%
8	DKK2	methylation	0.2168	4%
9	TG	expression	0.1978	6%
10	ZNF90	methylation	0.1515	2%
11	LOC619207	methylation	0.1479	12%
12	PTGFR	methylation	0.1445	2%
13	FCHSD1	methylation	0.1238	16%
14	SMN2	expression	0.1153	80%
15	KCNQ1OT1	methylation	0.1147	4%
16	MIR24-1	methylation	0.1113	8%
17	PAX6	expression	0.1107	100%
18	BCL2L10	expression	0.1103	4%
19	LGR5	expression	0.1089	2%
20	SLC44A4	expression	0.1075	42%
21	ST6GALNAC6	expression	0.1039	100%
22	DISP2	expression	0.1032	2%
23	ERAP2	expression	0.1032	100%
24	GRIK2	expression	0.1030	2%
25	RNASE6	methylation	0.1027	94%
26	CCL7	methylation	0.1015	34%
27	DDX4	expression	0.1009	2%
28	OR4K13	methylation	0.1000	2%
29	PLA2G4C	expression	0.0938	2%
30	CNTN2	expression	0.0916	6%

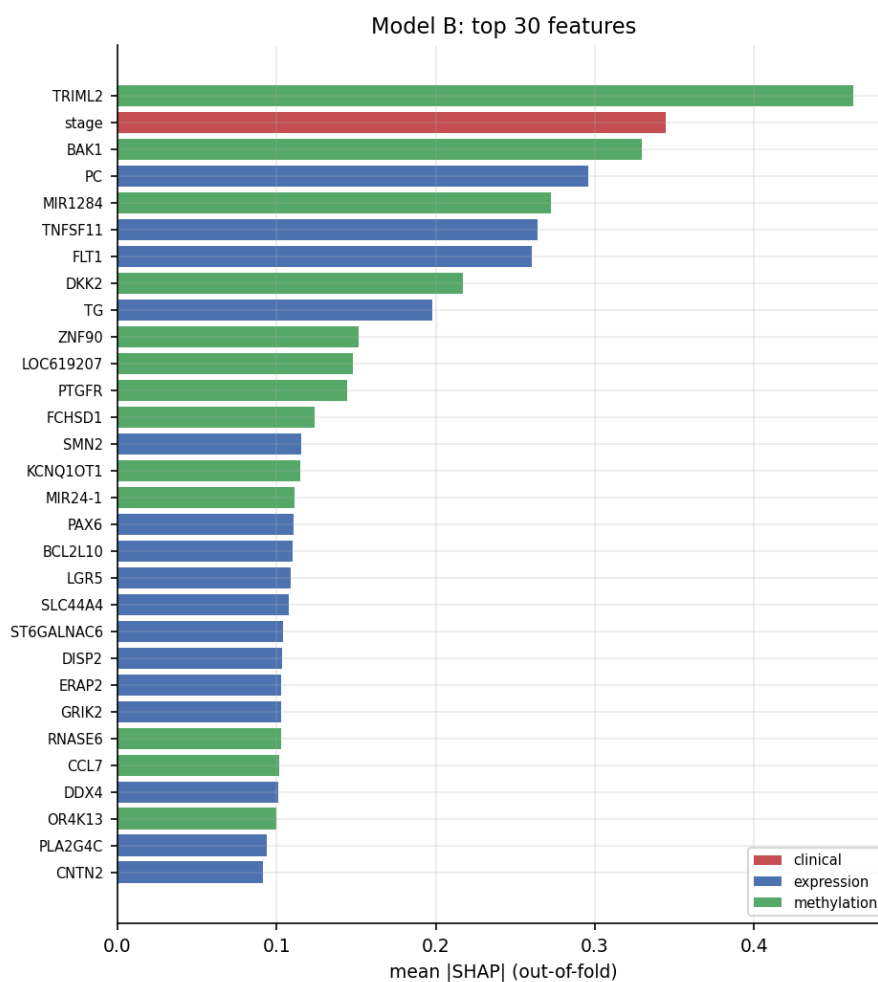


Figure 4. Model B top 30 features by out-of-fold mean |SHAP|.

4. Signature reduction (Gate 5)

Panel size	AUC (95% CI)	AP
20	0.662 [0.643, 0.681]	0.365
50	0.667 [0.645, 0.690]	0.359
100	0.683 [0.661, 0.705]	0.387
250	0.687 [0.666, 0.709]	0.389
500	0.682 [0.661, 0.704]	0.379

Performance plateaus at **100** markers (smallest panel within one standard error of the best mean AUC, which was 0.687 at 250 markers). 50-marker panel composition: {'expr': 31, 'meth': 17, 'clin': 2}.

Panels are re-selected inside every fold from the training half only; the reported marker list is the consensus across folds (selection frequency is given per marker in *results/tables/signature_panels.csv*). Scoring a globally-chosen panel would have leaked the test folds into the panel definition.

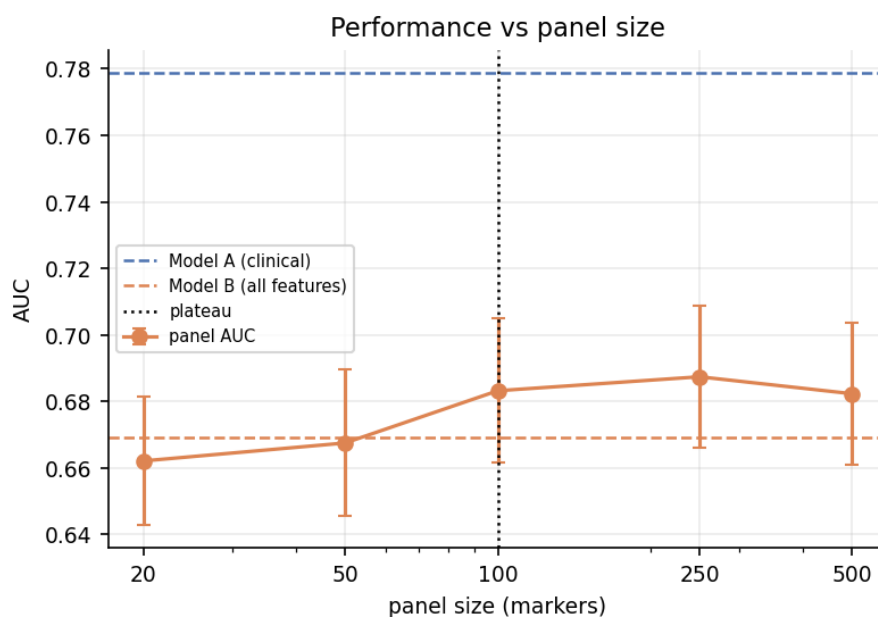


Figure 5. AUC against panel size, with model baselines.

4.1 The 50-marker panel

#	Marker	Block	Freq.	#	Marker	Block	Freq.
1	stage	clin	100%	26	TMEM45A	expr	38%
2	MIR1284	meth	100%	27	ANKRD55	expr	36%
3	FLT3	expr	88%	28	LOC100128842	meth	36%
4	BGLAP	meth	82%	29	INPP4B	expr	36%
5	RBM11	expr	82%	30	ZNF853	meth	36%
6	RUNDC3B	meth	82%	31	STC2	expr	36%
7	PLA2G10	expr	80%	32	LHX5	expr	34%
8	GSTM2	expr	72%	33	SYCP3	expr	34%
9	CHAD	expr	70%	34	MIR548F2	meth	34%
10	MBOAT1	expr	66%	35	ULBP2	expr	34%
11	MIR570	meth	66%	36	SDCBP2	expr	34%
12	PAX6	expr	62%	37	CLEC9A	expr	32%
13	BTG2	expr	62%	38	CCNG2	expr	32%

14	MS4A8	expr	60%	39	VGLL3	meth	30%
15	KIF5C	meth	58%	40	SNORA71A	meth	30%
16	NPB	meth	58%	41	PTGER3	expr	30%
17	GLI1	expr	56%	42	RLN1	expr	30%
18	CLIC6	expr	54%	43	AFF3	expr	30%
19	MIR612	meth	52%	44	ST6GALNAC6	expr	30%
20	SNORA67	meth	50%	45	MS4A18	expr	30%
21	age	clin	48%	46	RAI2	expr	28%
22	RNF133	meth	46%	47	CACNG4	expr	28%
23	ELOVL2	expr	40%	48	RP1	meth	28%
24	CHPT1	expr	40%	49	PGLYRP2	expr	26%
25	C2orf58	meth	38%	50	SNORD10	meth	26%

5. Immune deconvolution (Gate 6)

Method deviation, stated plainly. The spec asks for immunedeconv (R) with quanTIseq or MCP-counter. The immunedeconv install failed on this machine: Bioconductor dependencies (limSolve, biomaRt, sva, xCell, ConsensusTME, ComICS, quantiseqr) were unavailable. MCP-counter was therefore reimplemented directly in Python from the published 111-transcript, 10-population marker set (Becht et al., Genome Biology 2016) - the same signature the R package wraps. MCP-counter is a marker-gene aggregate, not a fitted deconvolution, so the reimplementaion is faithful rather than approximate. quanTIseq was NOT run: it requires the TIL10 signature matrix that ships with the R package.

Patients were split at the median out-of-fold risk score from Model B (AUC 0.669): 183 high-risk vs 184 low-risk. Observed relapse rate 27.9% vs 9.2%.

Population	High risk	Low risk	Delta	p	FDR
Endothelial cells	8.816	9.248	-0.432	7.46e-06	7.46e-05
Myeloid dendritic cells	5.734	6.382	-0.647	0.000259	0.00129
Neutrophils	5.544	5.838	-0.293	0.000725	0.00242
B lineage	5.819	6.368	-0.549	0.00762	0.0191
NK cells	2.770	3.045	-0.274	0.0147	0.0293
Fibroblasts	13.926	14.257	-0.331	0.0237	0.0395
T cells	6.393	6.710	-0.317	0.0664	0.0948
Cytotoxic lymphocytes	5.035	5.248	-0.213	0.105	0.131
Monocytic lineage	9.755	9.566	+0.189	0.157	0.175
CD8 T cells	6.782	6.930	-0.147	0.494	0.494

Gate 6 verdict. The signature separates risk groups that ALSO differ in immune infiltration -- it is at least partly an immune readout.

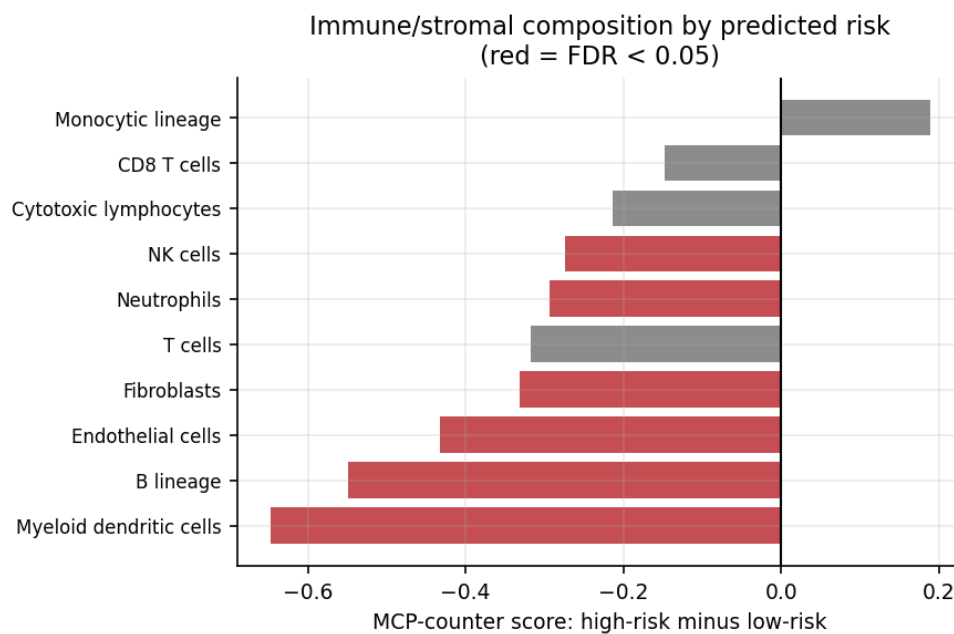


Figure 6. MCP-counter scores by predicted risk group.

6. External validation in METABRIC (Gate 7)

Known constraint, not worked around. METABRIC has no matched Illumina 450k methylation, so only the expression component of the signature can be tested externally. cBioPortal does host a promoter-RRBS methylation profile for METABRIC, but RRBS on a sample subset is a different assay and is not a substitute; it was not used. Transfer is also cross-platform (TCGA RNA-seq to Illumina HT-12 microarray), so genes were z-scored within each cohort independently, and degradation is expected.

Panel	Expr. markers	Mapped to METABRIC	Rate	Methylation markers (untestable)
20	9	9	100%	9
50	31	29	94%	17
100	66	62	94%	32
250	171	162	95%	77
500	345	319	92%	153

Panel	Model	Genes used	METABRIC AUC	METABRIC AP
20	logreg_l2	9	0.632	0.275
20	xgboost	9	0.558	0.217
50	logreg_l2	29	0.531	0.204
50	xgboost	29	0.558	0.228
100	logreg_l2	62	0.533	0.204
100	xgboost	62	0.579	0.226
250	logreg_l2	162	0.521	0.202
250	xgboost	162	0.585	0.234
500	logreg_l2	319	0.481	0.190
500	xgboost	319	0.568	0.217

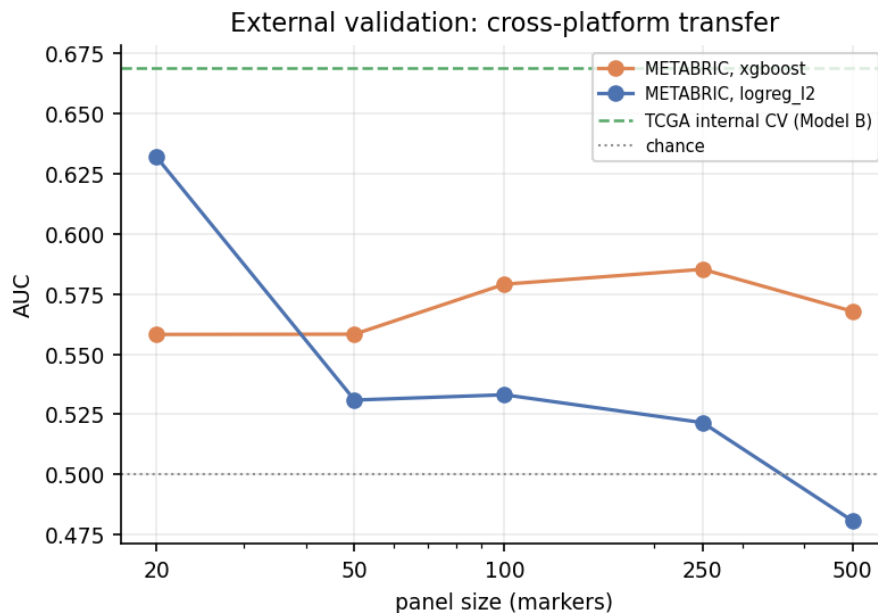


Figure 7. External validation across panel sizes.

Kaplan-Meier by predicted risk group. TCGA (out-of-fold risk): log-rank $p = 0.000179$; METABRIC (external): log-rank $p = 4.68e-06$.

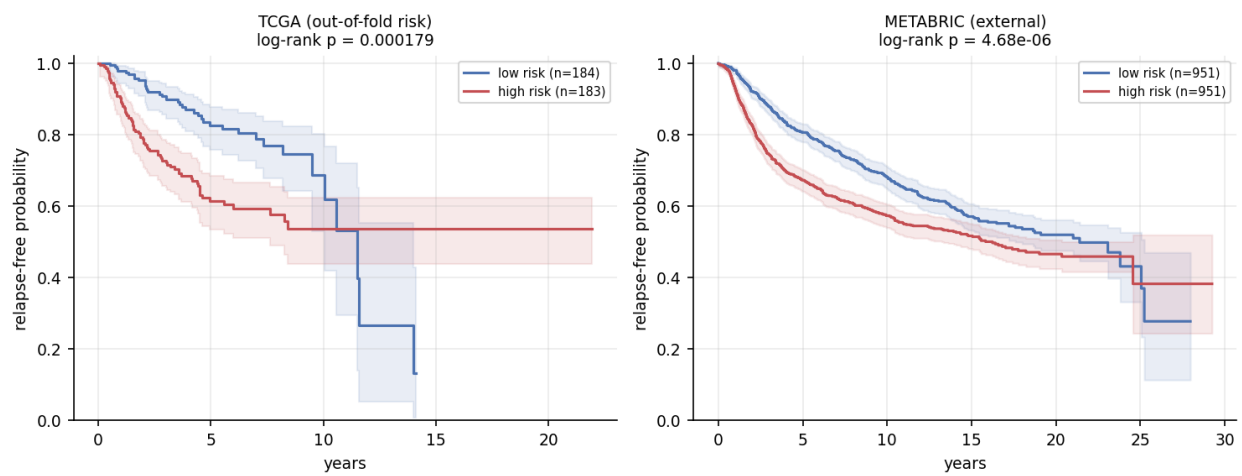


Figure 8. Relapse-free survival by predicted risk group. Note that negatives are required by the landmark rule to have full follow-up, which flattens the early part of the curves.

6b. Unsupervised multi-omics representation learning

The supervised pipeline is bottlenecked by $n=367$, because the landmark rule discards every patient censored before the horizon. An autoencoder needs no labels, so it trains on every patient with both assays: **777 patients**, 410 more than the labelled cohort. A denoising multi-modal autoencoder (separate encoders per modality, 4000 highest-variance features each, fused to **30 shared latent factors**, two decoders) was trained with early stopping on a held-out reconstruction split. No labels enter the encoder at any point.

Representation	AUC (95% CI)	AP
clinical only	0.778 [0.761, 0.794]	0.543
AE latent (transductive)	0.664 [0.641, 0.687]	0.367
AE latent + clinical (transductive)	0.781 [0.762, 0.800]	0.550
AE latent + clinical (transductive, xgb)	0.724 [0.703, 0.745]	0.445
AE latent (fold-safe)	0.670 [0.604, 0.736]	0.360
AE latent + clinical (fold-safe)	0.791 [0.732, 0.851]	0.561

Transduction check. Fitting one encoder on all patients and then cross-validating on its embeddings is transductive: the encoder has seen the FEATURES (never the labels) of patients who later appear in test folds. Reporting only that number would overstate the result, so the encoder was also refit inside each of 10 folds on training patients only, excluding that fold's test patients entirely. The gap is **-0.0046 AUC** - essentially zero, so the transductive result was not inflated.

What the factors encode. The latent factors correlate strongly with MCP-counter cell populations but barely with pathologic stage. Given no labels and no immune annotation, the autoencoder independently recovered the same immune/stromal axis that Gate 6 found by a different route - convergent evidence the axis is real, and an explanation for why these factors add little on top of stage: they encode a different, weaker signal.

Factor	r immune	Top population	r stage	Univariate AUC
21	0.43	Endothelial cells	0.05	0.634
15	0.26	CD8 T cells	0.00	0.615
5	0.41	Endothelial cells	0.10	0.589
25	0.37	Monocytic lineage	0.12	0.586
22	0.19	NK cells	0.03	0.584
12	0.23	Neutrophils	0.09	0.584
17	0.46	Fibroblasts	0.01	0.578
7	0.19	Neutrophils	0.03	0.577
14	0.32	CD8 T cells	0.09	0.572
20	0.37	Endothelial cells	0.11	0.568

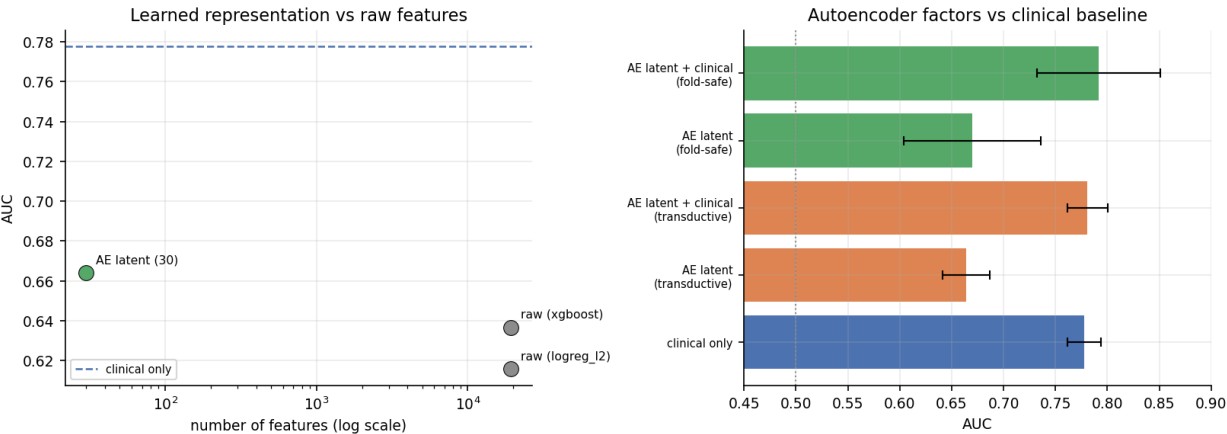


Figure 9. Learned latent factors versus raw features, and the transductive/fold-safe comparison.

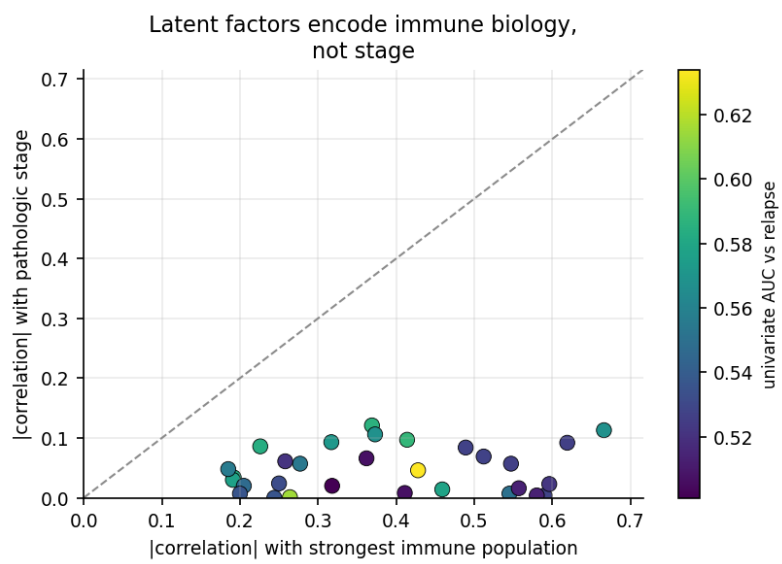


Figure 10. Latent factors track immune composition, not stage.

6c. Does each block earn its place? (ablation)

Feature block	Model	Features	AUC (95% CI)	AP
clinical	logreg_l2	7	0.778 [0.761, 0.794]	0.543
clinical	xgboost	7	0.742 [0.725, 0.759]	0.491
expression	logreg_l2	9758	0.614 [0.594, 0.634]	0.296
expression	xgboost	9758	0.638 [0.618, 0.658]	0.318
methylation	logreg_l2	9573	0.610 [0.587, 0.632]	0.290
methylation	xgboost	9573	0.595 [0.573, 0.617]	0.274
expression+methylation	logreg_l2	19331	0.616 [0.596, 0.636]	0.303
expression+methylation	xgboost	19331	0.636 [0.617, 0.656]	0.313
clinical+expression	logreg_l2	9765	0.647 [0.629, 0.666]	0.329
clinical+expression	xgboost	9765	0.676 [0.654, 0.697]	0.377
clinical+methylation	logreg_l2	9580	0.659 [0.637, 0.682]	0.333
clinical+methylation	xgboost	9580	0.643 [0.620, 0.666]	0.333
clinical+expression+methylation	logreg_l2	19338	0.645 [0.626, 0.664]	0.328
clinical+expression+methylation	xgboost	19338	0.669 [0.649, 0.690]	0.358

Paired contrasts on identical folds. Clinical covariates are forced into every combined model, so a 'clinical + omics' contrast cannot silently drop the clinical variables.

Question	Model	dAUC	p
does methylation add to expression?	logreg_l2	+0.0015	0.823
does methylation add to expression?	xgboost	-0.0013	0.85
does expression add to clinical?	logreg_l2	-0.1303	7.81e-16
does expression add to clinical?	xgboost	-0.0665	1.85e-05
does methylation add to clinical?	logreg_l2	-0.1183	3.53e-14
does methylation add to clinical?	xgboost	-0.0992	1.39e-09
does ANY omics add to clinical?	logreg_l2	-0.1327	3.08e-16
does ANY omics add to clinical?	xgboost	-0.0727	2.5e-07

6d. Does a LOCKED threshold transfer?

Every risk group elsewhere in this report came from a median split computed within each cohort, which is not how a deployed test works. Here a cutpoint is frozen in TCGA and applied unchanged to METABRIC.

Split	n high	n low	Relapse high	Relapse low	log-rank p
locked tcga cutpoint	1072	830	19.7%	17.8%	0.745
metabric own median	951	951	19.9%	17.9%	0.818

7. Limitations

- **Sample size.** $n = 367$ with ~19,000 candidate features. This is the dominant limitation and the most likely reason the omics models do not beat the clinical baseline. The spec's $n = 550-750$ target is not reachable in TCGA-BRCA once matched methylation and the early-censoring rule are both enforced.
- **Early-censoring exclusion.** 586 patients were discarded because their status at the horizon is genuinely unknown. This is the methodologically correct choice, but it induces a selection effect: negatives are, by construction, patients with long follow-up. Prevalence and the strength of stage as a predictor are both inflated relative to an unselected cohort.
- **No matched external methylation.** Only the expression half of the signature was externally validated. A multi-omics signature validated on one modality is a partially validated signature.
- **Cross-platform transfer.** RNA-seq to microarray transfer degrades performance independently of biology; the METABRIC number confounds signature quality with platform shift.
- **PAM50 in the 'clinical' baseline.** PAM50 is expression-derived. It is included because the spec lists it, but it makes Model A not a purely clinical model. The diagnostic in section 2.2 quantifies its (small) contribution.
- **Grade unavailable.** Histologic grade is 100% missing in the TCGA-CDR fields for this cohort and was dropped, so the clinical baseline is thinner than specified.
- **quantlseq not run**, and immunedeconv (R) not used - see section 5.
- **Single cohort discovery.** Discovery is entirely within TCGA-BRCA; no multi-cohort discovery or meta-analysis was attempted.

8. Reproducibility

Item	Value
Random seed	42
Outer CV	5-fold x 10 repeats (50 folds), shared across models via cv_folds.json
Inner CV	3-fold, 10 random search candidates
Leakage control	All selection/imputation/scaling inside sklearn Pipelines, refit per split
Container	docker/Dockerfile (python:3.11-slim, pinned)
Workflow	nextflow/main.nf (DSL2), profiles: standard, docker, conda, smoke
Smoke test	scripts/smoke_test.sh - full pipeline on a 100-patient fixture
Environment	env/requirements.txt (pip freeze), env/requirements-docker.txt

Every script takes `--config`, writes to `results/`, logs each filter with in/out counts, and records the seed. Figures are generated by a separate script that only reads tables, so a figure cannot disagree with the table behind it.